

VERIFIED EXECUTABLE LINEAR ALGEBRA

DenseMatrix makes Lean matrix algorithms run like data.

Row-major data for RREF, LU, and determinant kernels, bridged back to proof-friendly Matrix theorems.

TEAM

Annis Wu · Joseph Qian · Junye Ji
· Veer Shukla

Mentor: Dhruv Bhatia



Why DenseMatrix

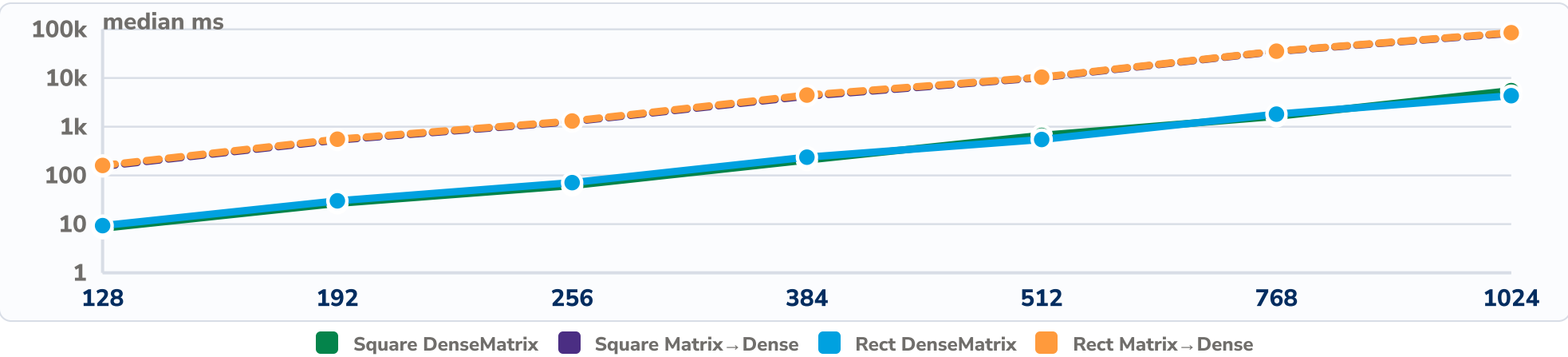
Row-major arrays let kernels work over contiguous data instead of re-evaluating pointwise Matrix functions.



Materialized executable workloads only; Matrix remains the proof API.

Multiplication Scaling

DenseMatrix vs Matrix → Dense, multiplication median ms.



15.1-21.5x faster

Benchmark Scope

2247	328	2048
RECORDS	MATERIALIZED CASES	MAX ROWS
Protocol30 repeats · 5 warmups		
Sizes16, 32, 64, 96, 128, 192, 256		
Stress384, 512, 768, 1024		
Machinepinned CPU core · archived environment		

Proof Connection

- DenseMatrix.toMatrix_add
- DenseMatrix.toMatrix_mul
- rowEchelonForm_toMatrix
- luDet_eq_det
- gaussDet_eq_det

Theorems connect executable kernels back to the Matrix API.

Representation Tradeoff

Median ms for pointwise primitives at n=1024.

OPERATION	DENSE	MATRIX	MATRIX → DENSE	FASTEST
Add	54.0	45.1	47.5	Matrix
Scalar	53.2	30.3	35.3	Matrix
Transpose	55.3	43.5	46.8	Matrix

Algorithm Kernels

Representative rational kernels at n=128.

REF	16,292 ms
RREF	32,394 ms
LU factor	17,090 ms
Gauss det	16,234 ms
LU det	16,933 ms